



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO RANGANAYAGI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

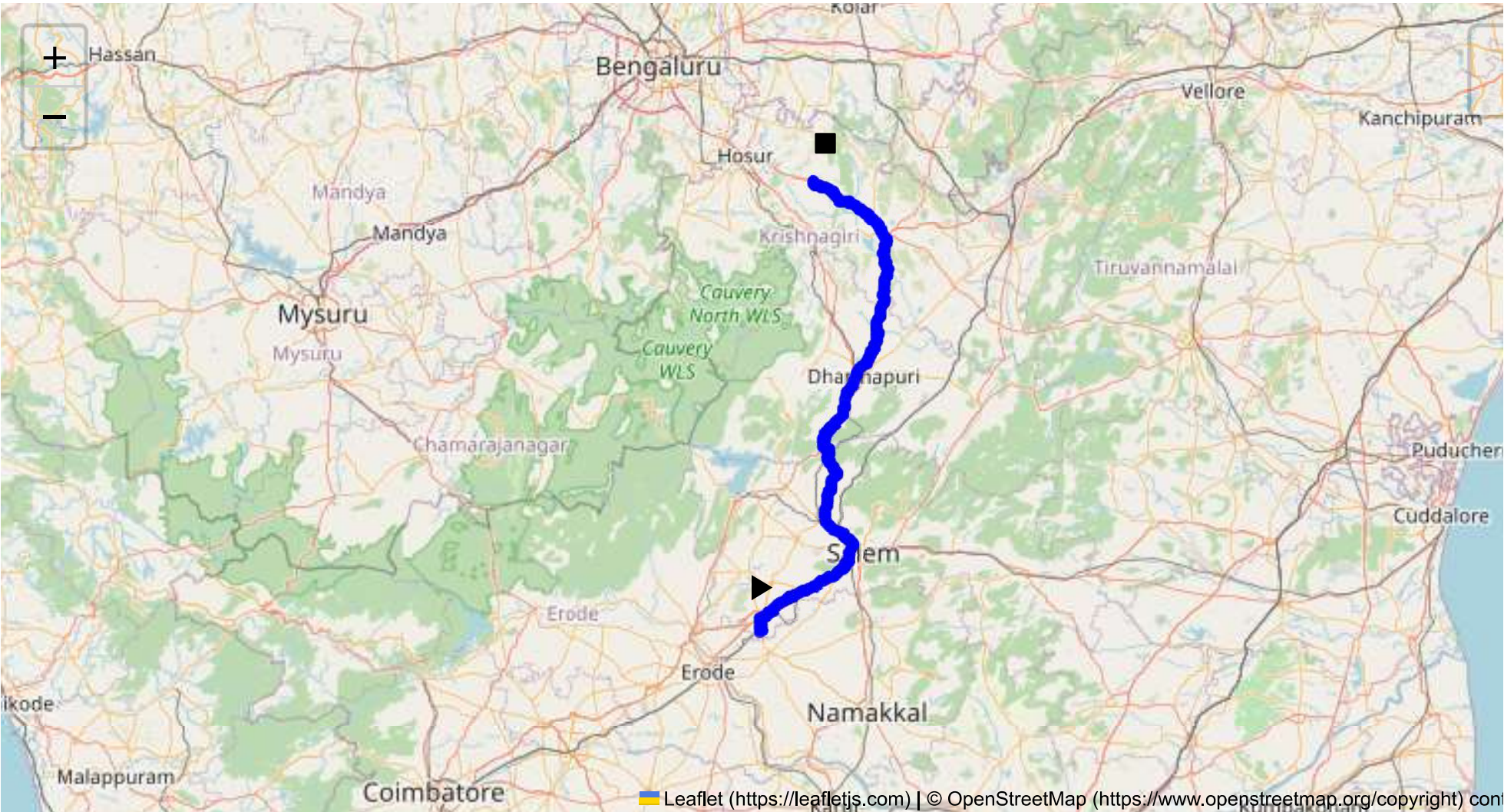
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 183.28 km
Estimated Duration: 3.3 hours
Adjusted Duration (Heavy Vehicle): 4.1 hours
Start: (11.4381, 77.8734)
End: (12.646497, 78.046389)



Welcome to the Journey Risk Management Study

Route Safety Analysis

1. Overview of the Route Map

The route from CVQF+23W in Sangagiri to J2WW+HH8 in Tamil Nadu covers approximately 183 kilometers. This journey traverses through rural and semi-urban areas, predominantly following major highways which are capable of accommodating heavy trucks. The route will likely include stretches on regional highways such as the NH44, which is well-maintained and frequently used by commercial vehicles.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu typically experiences tropical weather, with very hot summers, monsoon rains from October to December, and mild winters. Key weather-related hazards include:

- **Monsoon Rains:** Can cause water-logging, especially in low-lying areas, leading to reduced traction and visibility.
- **Seasonal Fog:** Possible during winter mornings, affecting visibility. Truck drivers should be cautious during the monsoon and early winter months.

3. Traffic Patterns

Traffic congestion tends to peak during:

- **Morning (7-10 AM) and Evening (5-8 PM):** Near urban areas and towns.
- **Congestion-Prone Areas:** Towns like Salem can be bottlenecks; planning alternate routes or off-peak travel times can mitigate delays.

4. Assessment of Road Quality and Infrastructure

- **Highways (e.g., NH44):** Generally good infrastructure with wide lanes, sufficient signage, and regular maintenance.
- **Rural Roads:** May have varying conditions, with potential for potholes and limited lighting.

5. Suggestions for Alternative Routes for Emergencies

- A backup route running parallel to NH44 via interior state highways could be considered, staying informed of local traffic alerts for any blockages.
- GPS-based real-time traffic updates can offer dynamic alternatives in case of road closures.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Must follow specific time windows set by local authorities for transporting hazardous materials.
- Obey signage for restricted areas and adhere to speed limits specifically for heavy vehicles.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- There have been periodic incidents linked to faulty vehicle maintenance and driver error.

- NH44 sees periodic accidents, underscoring the importance of defensive driving and adherence to road safety norms.

8. Environmental Considerations and Sensitive Areas

- The route may pass near agricultural lands; chemical leakage could impact soil and water.
- Near wildlife areas, mandatory measures should be taken to prevent harm to local fauna.

9. Analysis of Communication Coverage

- Mobile and internet coverage is robust along major highways due to good infrastructure but may drop in remote areas with fewer towers or dense vegetation.
- Satellite phones or portable communication devices are recommended for emergency use.

10. Estimated Emergency Response Times

- Urban areas: Swift response within 20-30 minutes due to proximity of facilities.
- Rural stretches: May take up to an hour or more, dependent on distance from emergency services.

11. Overall Summary of Risk Assessment

This route presents moderate risk, primarily tied to weather conditions and rural road segments. With proactive visibility and adherence to safety protocols, it is manageable. Emphasis should be on pre-trip preparation for inclement weather, staying informed on traffic updates, and ensuring vehicle integrity and communication readiness. Emergency response times vary based on geography, but the overall infrastructure supports efficient heavy vehicle transportation.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.46400, 77.87323	30 KM/Hr
5	Turn	Medium	11.46765, 77.87037	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	Medium	11.47986, 77.88397	30 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.53512, 77.95717	30 KM/Hr
11	Turn	Medium	11.53606, 77.95869	30 KM/Hr
12	Turn	Medium	11.62403, 78.10893	30 KM/Hr
13	Turn	Medium	11.62932, 78.11659	30 KM/Hr
14	Turn	High	11.64183, 78.11810	15 KM/Hr
15	Turn	Medium	11.64338, 78.11998	30 KM/Hr
16	Turn	Medium	11.67903, 78.12796	30 KM/Hr
17	Turn	Medium	11.69532, 78.10764	30 KM/Hr
18	Turn	Medium	11.73968, 78.05098	30 KM/Hr
19	Turn	Medium	11.86790, 78.08347	30 KM/Hr
20	Turn	Medium	11.91357, 78.05868	30 KM/Hr
21	Turn	Medium	11.92556, 78.06499	30 KM/Hr
22	Turn	Medium	11.93296, 78.06432	30 KM/Hr
23	Turn	Medium	11.93535, 78.06150	30 KM/Hr
24	Turn	Medium	11.95531, 78.04698	30 KM/Hr
25	Turn	High	11.95803, 78.04964	15 KM/Hr
26	Turn	Medium	11.96068, 78.04960	30 KM/Hr
27	Turn	High	11.96370, 78.05142	15 KM/Hr
28	Turn	High	11.96542, 78.04938	15 KM/Hr
29	Turn	Medium	11.96650, 78.04927	30 KM/Hr
30	Turn	Medium	11.96890, 78.05109	30 KM/Hr
31	Turn	Medium	12.22422, 78.19127	30 KM/Hr
32	Turn	Medium	12.24685, 78.19561	30 KM/Hr
33	Turn	Medium	12.25012, 78.19946	30 KM/Hr
34	Turn	Medium	12.25280, 78.20041	30 KM/Hr
35	Turn	Medium	12.29065, 78.19892	30 KM/Hr
36	Turn	Medium	12.32012, 78.20734	30 KM/Hr
37	Turn	Medium	12.40099, 78.21805	30 KM/Hr
38	Turn	Medium	12.45136, 78.21799	30 KM/Hr
39	Turn	Medium	12.50485, 78.22336	30 KM/Hr
40	Turn	Medium	12.59710, 78.13638	30 KM/Hr
41	Turn	Medium	12.61895, 78.08313	30 KM/Hr
42	Turn	Medium	12.63692, 78.07068	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
43	Blind Spot	Blind Spot	12.66002, 78.01673	10 KM/Hr
44	Turn	High	12.66017, 78.01709	15 KM/Hr
45	Turn	Medium	12.64632, 78.04564	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
6	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
8	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
9	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
14	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
20	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
21	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
23	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
24	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
25	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
26	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
27	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
28	hospital	Global Medical Centre Jagirammapalayam	11.6894409, 78.1108847	30 km/h	Medium
29	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
33	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
34	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
37	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
38	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
39	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
41	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
46	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium
47	hospital	St. Mary's Hospital	12.5128555, 78.2186887	30 km/h	Medium
48	hospital	KVS Hospital	12.5277857, 78.2055049	30 km/h	Medium
49	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
50	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
51	clinic	Shree Ramachandra ENT Centre	12.5290096, 78.2037359	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
15	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
22	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
30	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium
31	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
32	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
35	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
36	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
40	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
42	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
43	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
44	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium